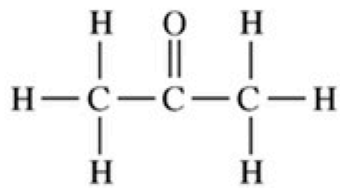
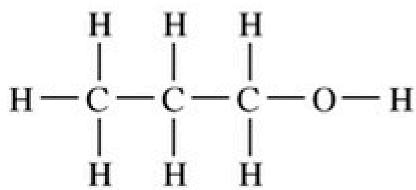
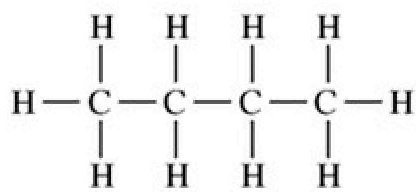
Chapter 03 Part 2: Intermolecular Forces

1.

Name	Structural Formula	Molar Mass (g/mol)
Acetone		58.1
1-propanol		60.1
Butane		58.1

The table above shows the structural formulas and molar masses for three different compounds. Which of the following is a list of the compounds in order of increasing boiling points?

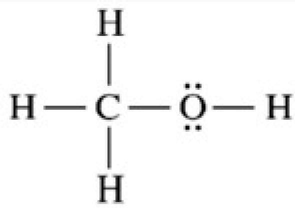
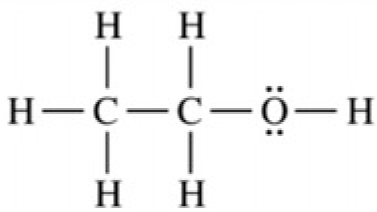
(A) Butane < 1-propanol < acetone

(B) Butane < acetone < 1-propanol

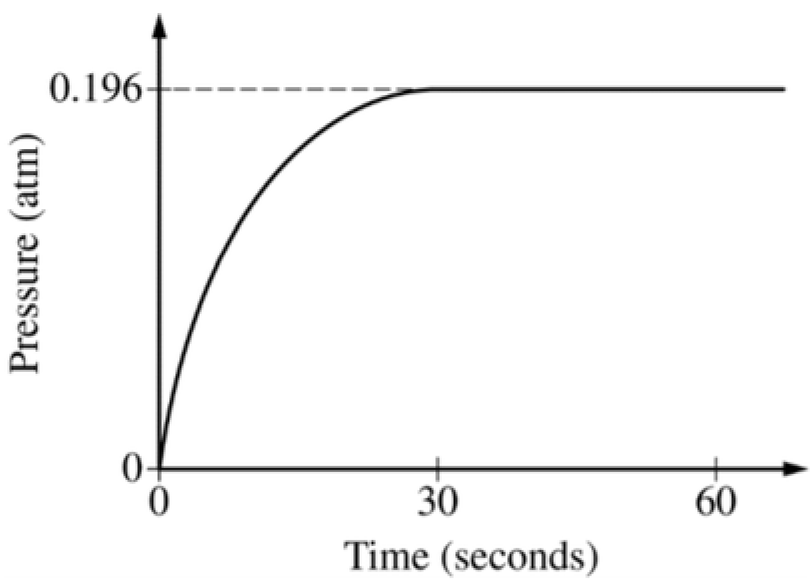
(C) 1-propanol < acetone < butane

(D) Acetone = butane < 1-propanol

Chapter 03 Part 2: Intermolecular Forces

Substance	Lewis Diagram	Boiling Point
CH_3OH		338 K
C_2H_5OH		351 K

Equimolar samples of $CH_3OH(l)$ and $C_2H_5OH(l)$ are placed in separate, previously evacuated, rigid 2.0 L vessels. Each vessel is attached to a pressure gauge, and the temperatures are kept at 300 K. In both vessels, liquid is observed to remain present at the bottom of the container at all times. The change in pressure inside the vessel containing $CH_3OH(l)$ is shown below.



2. Compared to the equilibrium vapor pressure of $CH_3OH(l)$ at 300 K, the equilibrium vapor pressure of $C_2H_5OH(l)$ at 300 K is

(A) The same, because both compounds have hydrogen bonding among their molecules

(B) higher, because London dispersion forces among C_2H_5OH molecules are greater than those among CH_3OH molecules

(C) lower, because London dispersion forces among C_2H_5OH molecules are greater than those among CH_3OH molecules ✓

(D) lower, because of the larger number of hydrogen bonds among C_2H_5OH molecules

---

3. Which statement best helps to explain the observation that $NH_3(l)$ boils at $-28^{\circ}C$, whereas $PH_3(l)$ boils at $-126^{\circ}C$?

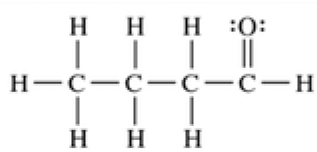
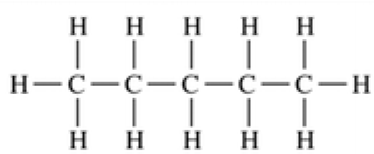
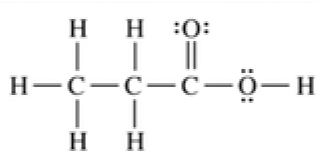
(A) The dispersion forces in NH_3 are weaker than the dispersion forces in PH_3.

(B) The dispersion forces in NH_3 are stronger than the dipole-dipole forces in PH_3.

(C) NH_3 has hydrogen bonding that is stronger than the dipole-dipole forces in PH_3. ✓

(D) NH_3 has hydrogen bonding that is weaker than the dipole-dipole forces in PH_3.

4.

Compound Name	Structural Formula	Molar Mass (g/mol)
Butanal		72.10
Pentane		72.15
Propanoic acid		74.08

Based on the information in the table above, which of the compounds has the highest boiling point, and why?

(A) Butanal, because it can form intermolecular hydrogen bonds

(B) Pentane, because it has the longest carbon chain

(C) Pentane, because it has the most $C-H$ bonds

(D) Propanoic acid, because it can form intermolecular hydrogen bonds ✓

Chapter 03 Part 2: Intermolecular Forces

5.

	Molar Mass (g/mol)	Boiling Point (°C)
$CS_2(l)$	76	46.5
$CCl_4(l)$	154	76.7

Based on the information in the table above, which liquid, $CS_2(l)$ or $CCl_4(l)$, has the higher equilibrium vapor pressure at 25°C, and why?

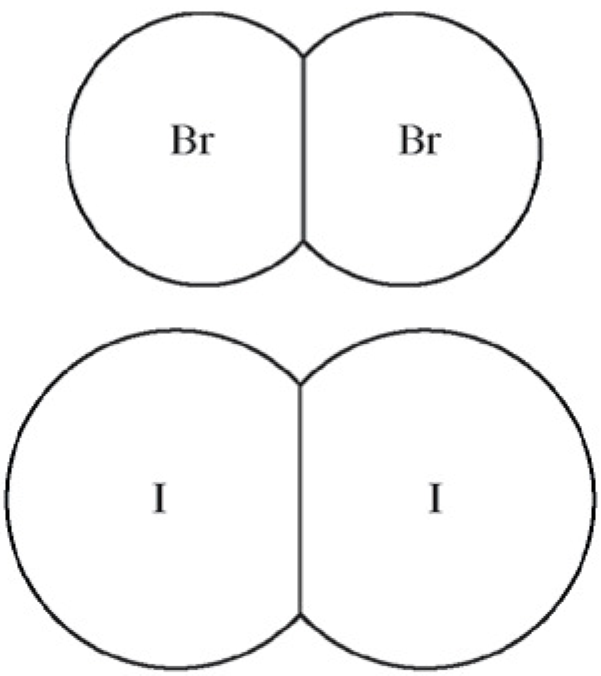
(A) $CS_2(l)$, because it has stronger London dispersion forces

(B) $CS_2(l)$, because it has weaker London dispersion forces

(C) $CCl_4(l)$, because it has stronger London dispersion forces

(D) $CCl_4(l)$, because it has weaker London dispersion forces

6.



The diagram above shows molecules of Br_2 and I_2 drawn to the same scale. Which of the following is the best explanation for the difference in the boiling points of liquid Br_2 and I_2, which are 59°C and 184°C, respectively?

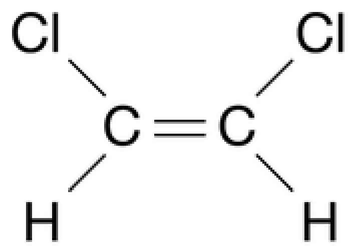
(A) Solid iodine is a network covalent solid, whereas solid bromine is a molecular solid.

(B) The covalent bonds in I_2 molecules are weaker than those in Br_2 molecules.

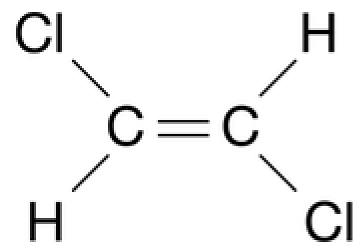
(C) I_2 molecules have electron clouds that are more polarizable than those of Br_2 molecules, thus London dispersion forces are stronger in liquid I_2. ✓

(D) Bromine has a greater electronegativity than iodine, thus there are stronger dipole-dipole forces in liquid bromine than in liquid iodine.

7.



cis-isomer



trans-isomer

The structural formulas for two isomers of 1, 2-dichloroethene are shown above. Which of the two liquids has the higher equilibrium vapor pressure at $20^\circ C$, and why?

(A) The *cis*-isomer, because it has dipole-dipole interactions, whereas the *trans*-isomer has only London dispersion forces

(B) The *cis*-isomer, because it has only London dispersion forces, whereas the *trans*-isomer also has dipole-dipole interactions

(C) The *trans*-isomer, because it has dipole-dipole interactions, whereas the *cis*-isomer has only London dispersion forces

(D) The *trans*-isomer, because it has only London dispersion forces, whereas the *cis*-isomer also has dipole-dipole interactions ✓

Chapter 03 Part 2: Intermolecular Forces

8.

Name	Molecular Formula	Molar Mass (g/mol)
Ethane	C_2H_6	30
Butane	C_4H_{10}	58

The molecular formula and molar mass of two straight-chain hydrocarbons are listed in the table above. Based on the information in the table, which compound has the higher boiling point, and why is that compound's boiling point higher?

(A) C_4H_{10} ,because it has more hydrogen atoms, resulting in more hydrogen bonding

(B) C_4H_{10} , because it has more electrons, resulting in greater polarizability and stronger dispersion forces ✓

(C) C_2H_6 , because its molecules are smaller and they can get closer to one another, resulting in stronger dispersion forces

(D) C_2H_6 , because its molecules are more polar, resulting in stronger dipole-dipole attractions

9. On the basis of strength of intermolecular forces, which of the following elements would be expected to have the highest melting point?

(A) Br_2 ✓

(B) Cl_2

(C) F_2

(D) Kr

(E) N_2

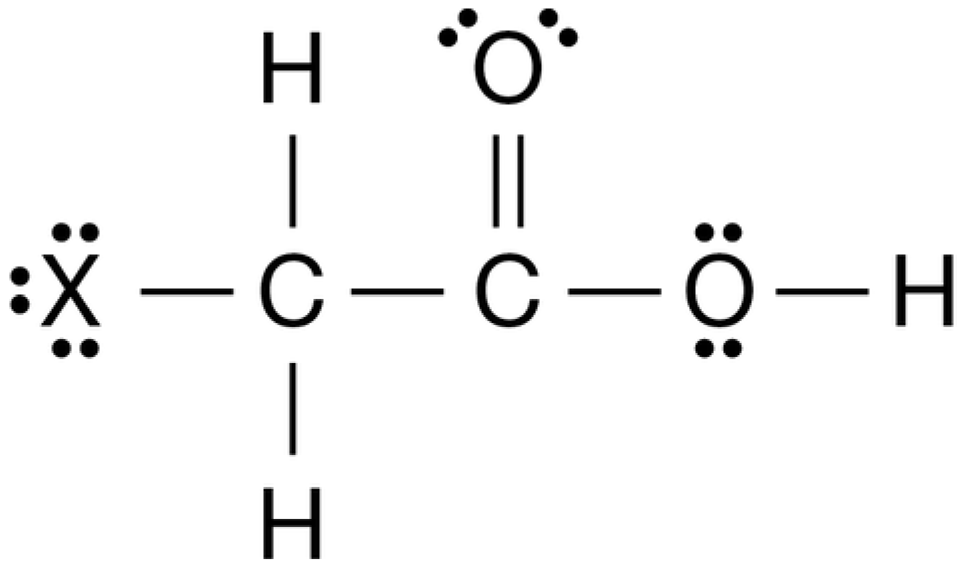
10.

Hydrogen Halide	Normal Boiling Point, ˚C
HF	+19
HCl	–85
HBr	–67
HI	–35

The liquefied hydrogen halides have the normal boiling points given above. The relatively high boiling point of HF can be correctly explained by which of the following?

Chapter 03 Part 2: Intermolecular Forces

(A) HF gas is more ideal.
(B) HF is the strongest acid.
(C) HF molecules have a smaller dipole moment.
(D) HF is much less soluble in water.
(E) HF molecules tend to form hydrogen bonds.



The structure of haloacetic acids, XCH_2COOH (where X is either F, Cl, Br, or I), is shown above. The dissociation constants and molar masses of four haloacetic acids are listed in the table below.

Acid	pK_a	K_a	**Molar Mass** (g/mol)
Fluoroacetic acid	2.59	2.57×10^{-3}	78.0
Chloroacetic acid	2.87	1.35×10^{-3}	94.5
Bromoacetic acid	2.90	1.26×10^{-3}	138.9
Iodoacetic acid	3.18	6.61×10^{-4}	185.9

11. Which compound, chloroacetic acid or iodoacetic acid, most likely has the lower boiling point, and why?

(A) Chloroacetic acid, because the London dispersion forces among its molecules are weaker.
(B) Chloroacetic acid, because the dipole-dipole forces among its molecules are weaker.
(C) Iodoacetic acid, because the London dispersion forces among its molecules are stronger.
(D) Iodoacetic acid, because the dipole-dipole forces among its molecules are stronger.

Chapter 03 Part 2: Intermolecular Forces

12.

Substance	Equilibrium Vapor Pressure at 20°C (torr)
$C_6H_6(l)$	75
$C_2H_5OH(l)$	44
$CH_3OH(l)$	92
$C_2H_6O_2(l)$	0.06

Based on the data in the table above, which of the following liquid substances has the weakest intermolecular forces?

(A) $C_6H_6(l)$

(B) $C_2H_5OH(l)$

(C) $CH_3OH(l)$ ✓

(D) $C_2H_6O_2(l)$

13. Which of the following best helps to explain why CCl_4 is a liquid whereas CI_4 is a solid when both are at 25°C?

(A) The dipole moment of the CCl_4 molecule is larger than that of the CI_4 molecule because Cl is more electronegative than I.

(B) The dipole moment of the CI_4 molecule is larger than that of the CCl_4 molecule because there is stronger repulsion between electrons in the C-I bonds compared to the repulsion between electrons in the C- Cl bonds.

(C) The London dispersion forces are stronger in CCl_4 than in CI_4 because Cl is more electronegative than I.

(D) The London dispersion forces are stronger in CI_4 than in CCl_4 because CI_4 has a more polarizable electron cloud than CCl_4 ✓

14. In solid methane, the forces between neighboring CH_4 molecules are best characterized as

(A) ionic bonds

(B) covalent bonds

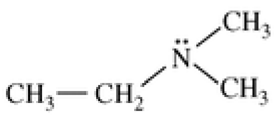
(C) hydrogen bonds

(D) ion-dipole forces

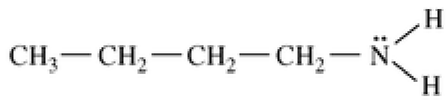
(E) London (dispersion) forces ✓

Chapter 03 Part 2: Intermolecular Forces

15.



Compound 1



Compound 2

Based on the structures shown above, which of the following statements identifies the compound with the higher boiling point and provides the best explanation for the higher boiling point?

(A) Compound 1, because it has stronger dipole-dipole forces than compound 2

(B) Compound 1, because it forms hydrogen bonds, whereas compound 2 does not

(C) Compound 2, because it is less polarizable and has weaker London dispersion forces than compound 1

(D) Compound 2, because it forms hydrogen bonds, whereas compound 1 does not

16. Ne, HF, C_2H_6, CH_4

Which of the substances listed above has the highest boiling point, and why?

(A) Ne, because its atoms have the largest radius

(B) HF, because its molecules form hydrogen bonds

(C) C_2H_6 , because each molecule can form multiple hydrogen bonds

(D) CH_4 , because its molecules have the greatest London dispersion forces

17. The boiling points of the elements helium, neon, argon, krypton, and xenon increase in that order. Which of the following statements accounts for this increase?

(A) The London (dispersion) forces increase.

(B) The hydrogen bonding increases.

(C) The dipole-dipole forces increase.

(D) The chemical reactivity increases.

(E) The number of nearest neighbors increases.

18. Which of the following lists the substances F_2, HCl, and HF in order of increasing boiling point?

(A) $HF < HCl < F_2$

(B) $HF < F_2 < HCl$

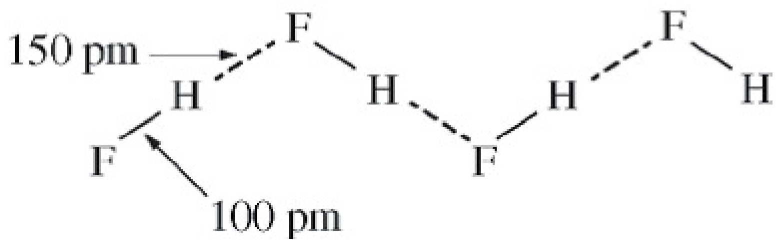
(C) $HCl < F_2 < HF$

(D) $HCl < HF < F_2$

(E) $F_2 < HCl < HF$

Chapter 03 Part 2: Intermolecular Forces

19.



The figure above shows that in solid hydrogen fluoride there are two different distances between H atoms and F atoms. Which of the following best accounts for the two different distances?

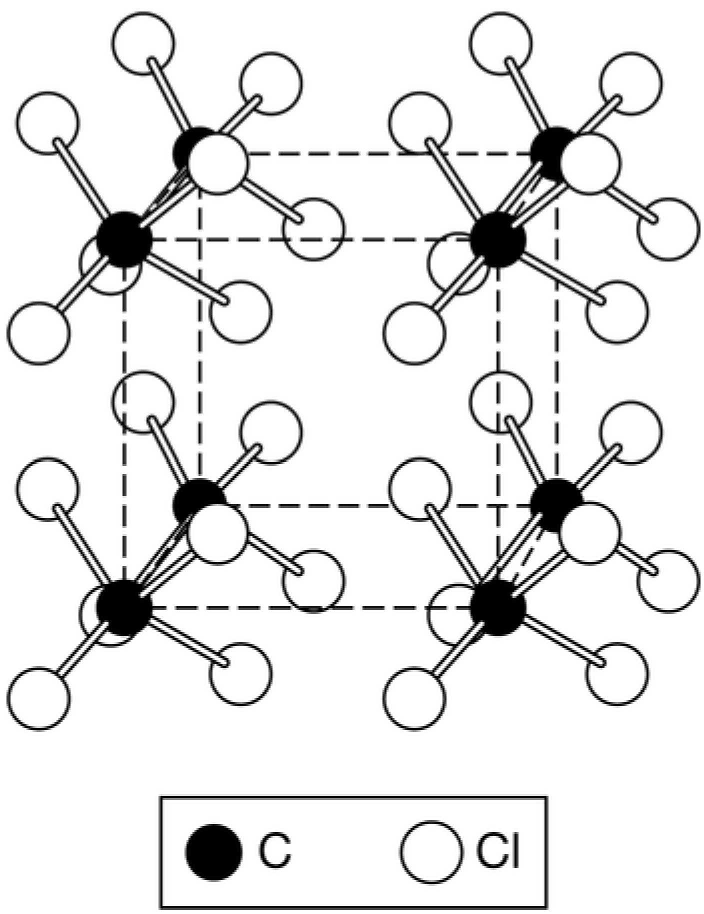
(A) Accommodation of the necessary bond angles in the formation of the solid

(B) Difference in strength between covalent bonds and intermolecular attractions ✓

(C) Different isotopes of fluorine present in the samples

(D) Uneven repulsions among nonbonding electron pairs

20.



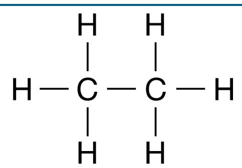
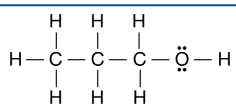
Solid carbon tetrachloride, $CCl_4(s)$, is represented by the diagram above. The attractions between the CCl_4 molecules that hold the molecules together in the solid state are best identified as

(A) polar covalent bonds

(B) nonpolar covalent bonds

(C) intermolecular attractions resulting from temporary dipoles ✓

(D) intermolecular attractions resulting from permanent dipoles

Compound	Molar Mass	Balanced Equation for the Combustion of One Mole of the Compound	$\Delta H^\circ{}_{comb}$ $(\mathrm{kJ/mol}_{rxn})$
 Ethane	$30.\,\mathrm{g/mol}$	$C_2H_6(g) + \frac{7}{2}O_2(g) \rightarrow 2\,CO_2(g) + 3\,H_2O(l)$	-1560
 Propanol	$60.\,\mathrm{g/mol}$	$C_3H_8O(l) + \frac{9}{2}O_2(g) \rightarrow 3\,CO_2(g) + 4\,H_2O(l)$	-2020
Unknown	?	$? + \ 5\,O_2(g) \rightarrow 4\,CO_2(g) + 4\,H_2O(l)$	-2240

21. Based on the structural formulas, which of the following identifies the compound that is more soluble in water and best helps to explain why?

(A) Ethane, because the electron clouds of its molecules are more polarizable than those of propanol.

(B) Ethane, because its molecules can fit into the spaces between water molecules more easily than those of propanol can.

(C) Propanol, because its molecules have a greater mass than the molecules of ethane have.

(D) Propanol, because its molecules can form hydrogen bonds with water molecules but those of ethane cannot. ✓

Refer to three gases in identical rigid containers under the conditions given in the table below.

Container	A	B	C
Gas	Methane	Ethane	Butane
Formula	CH_4	C_2H_6	C_4H_{10}
Molar mass (g/mol)	16	30.	58
Temperature (°C)	27	27	27
Pressure (atm)	2.0	4.0	2.0

22. If the pressure of each gas is increased at constant temperature until condensation occurs, which gas will condense at the lowest pressure?

(A) Methane

(B) Ethane

(C) Butane

(D) All the gases will condense at the same pressure.

---

23. The London (dispersion) forces are weakest for which of the following gases under the same conditions of temperature and pressure?

(A) H_2

(B) O_2

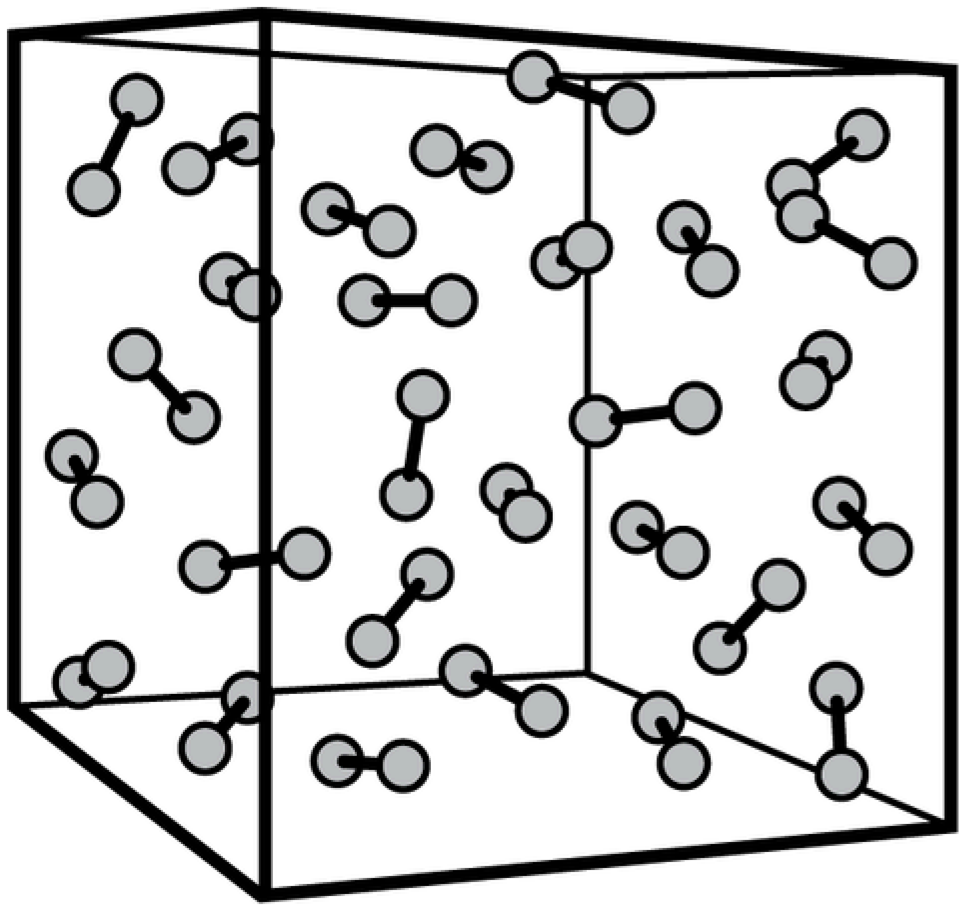
(C) Xe

(D) F_2

(E) N_2

Chapter 03 Part 2: Intermolecular Forces

24.



The diagram above is a molecular model of a gaseous diatomic element that is just above its boiling point. Intermolecular forces between the gas molecules will cause them to condense into the liquid phase if the temperature is lowered. Which of the following best describes how the model is limited in its depiction of the phenomenon?

(A) It does not show how hydrogen bonds are constantly forming, breaking, and reforming, which results in a net force of attraction between the molecules.

(B) It does not show how the interactions between ions and the induced molecular dipoles result in a net force of attraction between the molecules.

(C) It does not show how the interacting permanent dipoles of the molecules result in a net force of attraction between the molecules.

(D) It does not show how the temporary fluctuating dipoles of the molecular electron clouds result in a net force of attraction between the molecules. ✓

Chapter 03 Part 2: Intermolecular Forces

Directions: Each set of lettered choices below refers to the numbered statements immediately following it. Select the one lettered choice that best fits each statement. A choice may be used once, more than once, or not at all in each set.

(A) A network solid with covalent bonding
(B) A molecular solid with zero dipole moment
(C) A molecular solid with hydrogen bonding
(D) An ionic solid
(E) A metallic solid

25. Solid ethyl alcohol, C_2H_5OH

(A) A network solid with covalent bonding

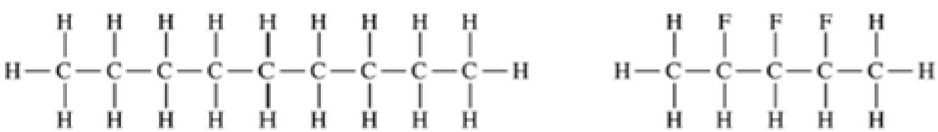
(B) A molecular solid with zero dipole moment

(C) A molecular solid with hydrogen bonding ✓

(D) An ionic solid

(E) A metallic solid

26.



Nonane

2,3,4-trifluoropentane

Consider the molecules represented above and the data in the table below.

Compound	Molecular Formula	Molar Mass (g/mol)	Boiling Point (°C)
Nonane	C_9H_{20}	128	151
2,3,4-trifluoropentane	$C_5H_9F_3$	126	89

Nonane and 2,3,4-trifluoropentane have almost identical molar masses, but nonane has a significantly higher boiling point. Which of the following statements best helps explain this observation?

(A) The C-F is easier to break than the C-H bond.

(B) The C-F is more polar than the C-H bond.

(C) The carbon chains are longer in nonane than they are in 2,3,4-trifluoropentane. ✓

(D) The carbon chains are farther apart in a sample of nonane than they are in 2,3,4-trifluoropentane.

Chapter 03 Part 2: Intermolecular Forces

27.

Ion	Ionic Radius (pm)
Li^{+}	60
Na^{+}	95
Ca^{2+}	99
In^{3+}	81

Based on Coulomb's Law and the information in the table above, which of the following cations is most likely to have the weakest interaction with an adjacent water molecule in an aqueous solution?

(A) Li^{+}

(B) Na^{+} ✓

(C) Ca^{2+}

(D) In^{3+}

28.

Substance	Normal boiling point
HF	293 K
F_2	85 K

The electron cloud of HF is smaller than that of F_2 , however, HF has a much higher boiling point than F_2 has. Which of the following explains how the dispersion-force model of intermolecular attraction does not account for the unusually high boiling point of HF?

(A) F_2 is soluble in water, whereas HF is insoluble in water.

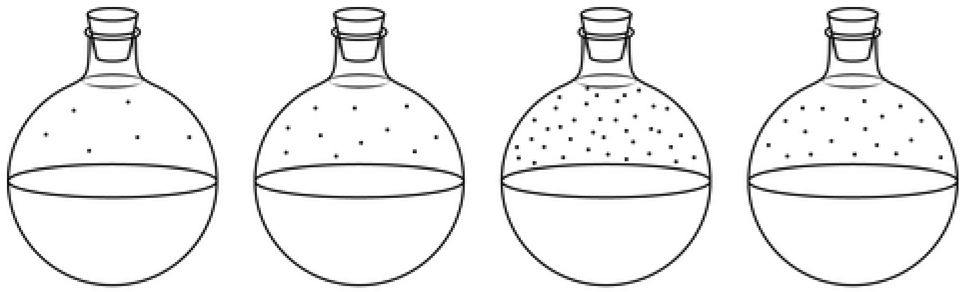
(B) The F_2 molecule has a greater mass than the HF molecule has.

(C) Liquid F_2 has weak dispersion force attractions between its molecules, whereas liquid HF has strong ionic interactions between H^{+} and F^{-} ions.

(D) Liquid F_2 has weak dispersion force attractions between its molecules, whereas liquid HF has both weak dispersion force attractions and hydrogen bonding interactions between its molecules. ✓

Chapter 03 Part 2: Intermolecular Forces

29.



Four different liquid compounds in flasks at $20^\circ C$ are represented above. The table below identifies the compounds. Flask C shows the most particles in the vapor phase. Which of the following is not shown in the model but best helps to explain why flask C must contain pentane?

Name	Chemical Formula	Boiling Point $(^\circ C)$
Pentane	$CH_3CH_2CH_2CH_2CH_3$	36
Hexane	$CH_3CH_2CH_2CH_2CH_2CH_3$	69
Heptane	$CH_3CH_2CH_2CH_2CH_2CH_2CH_3$	98
Octane	$CH_3CH_2CH_2CH_2CH_2CH_2CH_2CH_3$	126

(A) The random motion of the particles within the liquids

(B) The relative speeds of the vapor particles in each flask

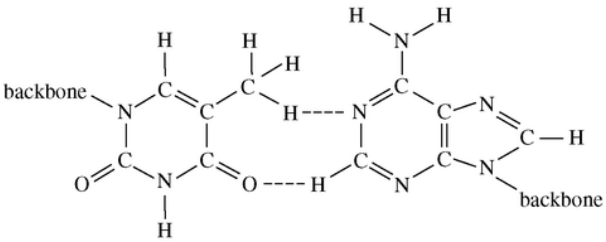
(C) The strength of the intermolecular forces between the particles in the liquids

(D) The structural formula of the molecules of the liquid and vapor in each flask

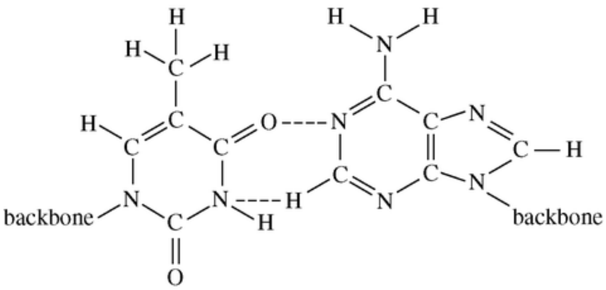
30. Thymine and adenine form a base pair in the DNA molecule. These two bases can form a connection between two strands of DNA via two hydrogen bonds. Which of the following diagrams shows the correct representation of the hydrogen bonding (denoted by dashed lines) between thymine and adenine base pairs? (In each diagram, thymine is shown at the left and adenine is shown at the right. The bases are attached to the backbone portion of the DNA strands.)

Chapter 03 Part 2: Intermolecular Forces

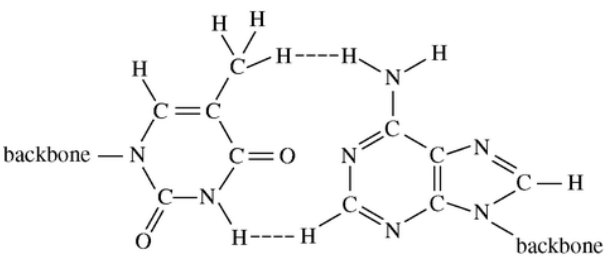
(A)



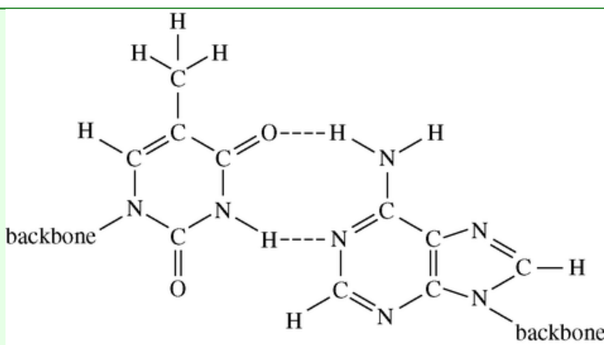
(B)



(C)



(D) ✓



Chapter 03 Part 2: Intermolecular Forces

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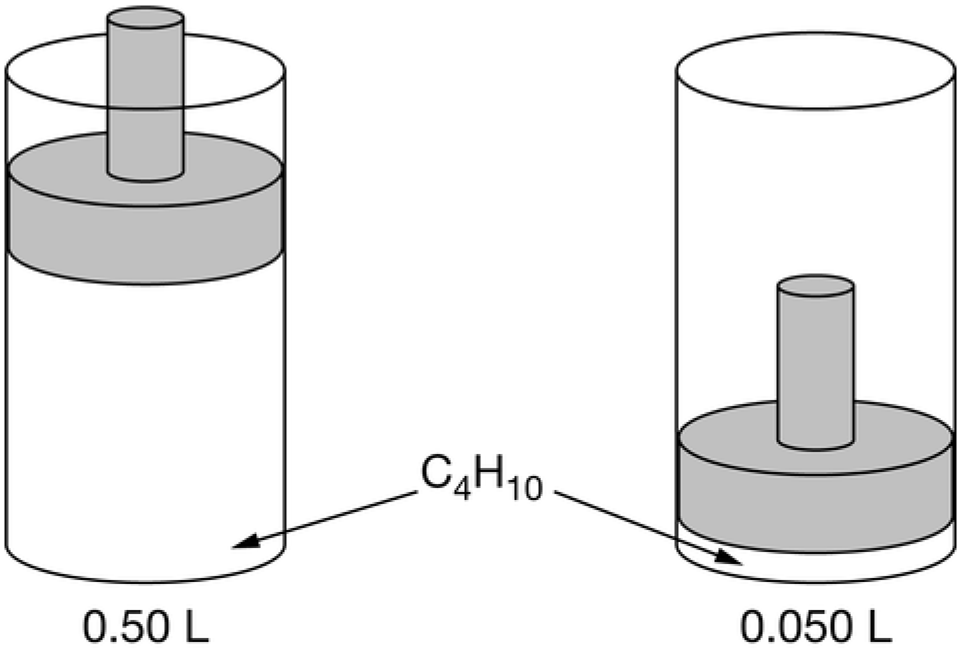
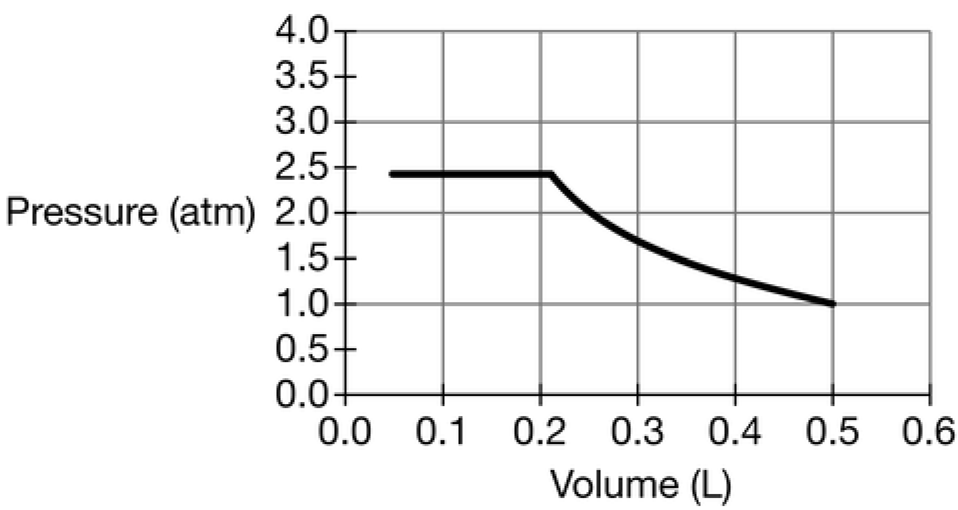
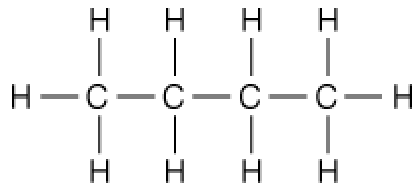


A sample of $C_4H_{10}(g)$ is placed in a rigid vessel with a movable piston. The volume of the gas is reduced by moving the piston downward while the temperature is maintained at $25^\circ C$, as shown above. The pressure of the gas versus its volume is plotted in the graph below.



Chapter 03 Part 2: Intermolecular Forces

31.



The Lewis electron-dot diagram for C_4H_{10} is shown above. Which of the following is the predominant type of intermolecular force among C_4H_{10} molecules?

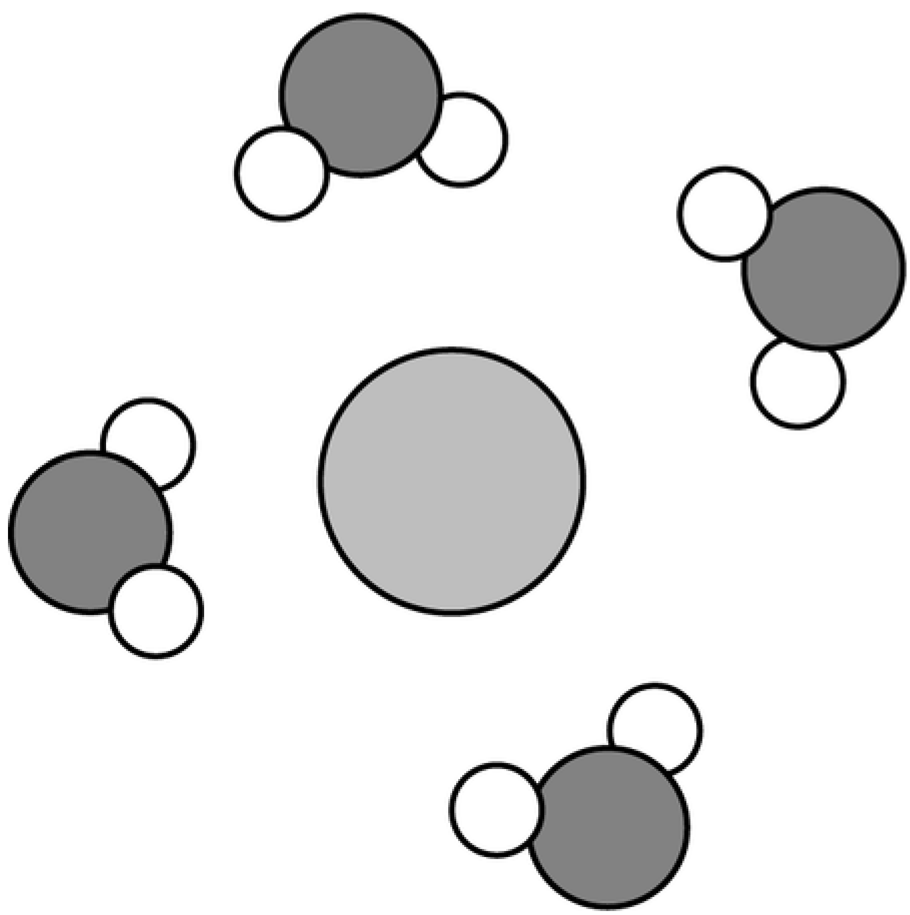
(A) Hydrogen bonding

(B) London dispersion forces ✓

(C) Dipole-dipole attractions

(D) Dipole-induced dipole attractions

32.



The diagram above represents a particle in aqueous solution. Which of the following statements about the particle is correct?

Chapter 03 Part 2: Intermolecular Forces

(A) The particle must be a cation because the negative end of each water molecule is pointed toward it.

(B) The particle must be an anion because the positive end of each water molecule is pointed toward it. ✓

(C) The charge of the particle cannot be determined because water molecules have no net charge.

(D) The charge of the particle cannot be determined because the water molecules are arranged symmetrically and their partial charges cancel.

33.

Ion	Ionic Radius (pm)
Cl^-	181
I^-	216
S^{2-}	184
Te^{2-}	221

Based on Coulomb's law and the information in the table above, which of the following anions is most likely to have the strongest interactions with nearby water molecules in an aqueous solution?

(A) Cl^-

(B) I^-

(C) S^{2-} ✓

(D) Te^{2-}

34. Which element, Ne or Xe, has the lower boiling point, and why?

(A) Ne, because it has a smaller, less polarizable electron cloud than Xe ✓

(B) Ne, because its atoms have a faster average speed at a given temperature than Xe atoms

(C) Xe, because it has a larger, more polarizable electron cloud than Ne

(D) Xe, because its atoms have a slower average speed at a given temperature than Ne atoms

35. At $298\ K$ and $1\ atm$, Br_2 is a liquid and I_2 is a crystalline solid. Which of the following statements best explains the different physical states of Br_2 and I_2 at $298\ K$ and $1\ atm$?

(A) The London dispersion forces among I_2 molecules are stronger than those among Br_2 molecules. ✓

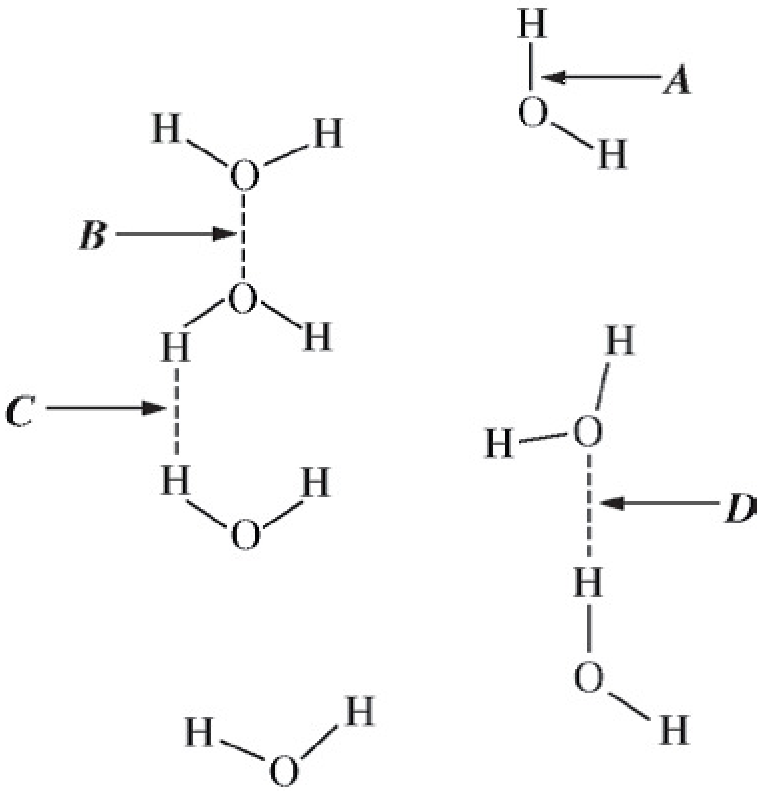
(B) The London dispersion forces among Br_2 molecules are stronger than those among I_2 molecules.

(C) The $Br-Br$ bond is stronger than the $I-I$ bond.

(D) The $I-I$ bond is stronger than the $Br-Br$ bond.

Chapter 03 Part 2: Intermolecular Forces

36.



In the diagram above, which of the labeled arrows identifies hydrogen bonding in water?

(A) *A*

(B) *B*

(C) *C*

(D) *D*

37.

Ion	Radius (pm)
Zn^{2+}	74
Ca^{2+}	100
Ba^{2+}	135

Based on the data in the table above, which of the following correctly predicts the relative strength of the attraction of Zn^{2+}, Ca^{2+}, and Ba^{2+} ions to water molecules in a solution, from strongest to weakest, and provides the correct reason?

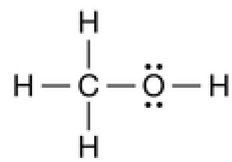
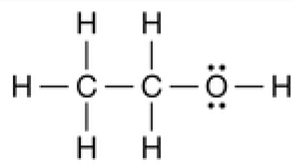
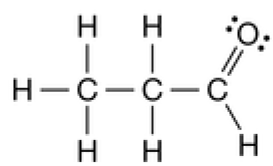
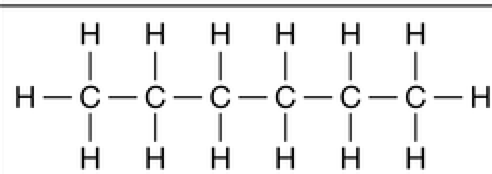
Chapter 03 Part 2: Intermolecular Forces

(A) $Zn^{2+} > Ca^{2+} > Ba^{2+}$ because the smaller ions have a stronger coulombic attraction to water

(B) $Zn^{2+} > Ca^{2+} > Ba^{2+}$ because the smaller ions are more electronegative

(C) $Ba^{2+} > Ca^{2+} > Zn^{2+}$ because the larger ions are more polarizable

(D) $Ba^{2+} > Ca^{2+} > Zn^{2+}$ because the larger ions are less electronegative

Compound	Structure	Boiling Point (°C)
Methanol		65
Ethanol		78
Propanal		49
Hexane		69

38. Which of the following best helps to explain why hexane has a higher boiling point than methanol has?

(A) Methanol molecules can form hydrogen bonds with other methanol molecules.

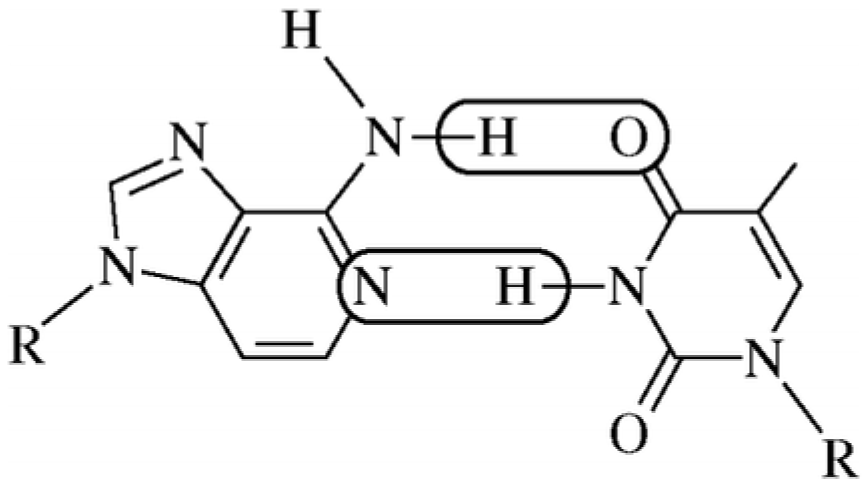
(B) Hexane cannot form hydrogen bonds with other hexane molecules.

(C) Methanol molecules have attractions to one another due to London dispersion forces.

(D) Hexane molecules have electron clouds that are larger than those of methanol molecules.

Chapter 03 Part 2: Intermolecular Forces

39.



Which of the following is the strongest type of interaction that occurs between the atoms within the circled areas of the two molecules represented above?

(A) Polar covalent bond

(B) Nonpolar covalent bond

(C) Hydrogen bond ✓

(D) London dispersion forces

40. Use the principles of atomic structure and/or chemical bonding to explain each of the following. In each part, your answers must include references to **both** substances.

a. The atomic radius of Li is larger than that of Be.

b. The second ionization energy of K is greater than the second ionization energy of Ca.

c. The carbon-to-carbon bond energy in C_2H_4 is greater than it is in C_2H_6.

d. The boiling point of Cl_2 is lower than the boiling point of Br_2.

Part A

1 point is earned for indicating that Be has more protons than Li

1 point is earned for indicating that since the electrons are at about the same distance from the nucleus, there is more attraction in Be as a result of the larger number of protons

Both Li and Be have their outer electrons in the same shell (and/or they have the same number of inner core electrons shielding the valence electrons from the nucleus). However, Be has four protons and Li has only three protons. Therefore, the effective nuclear charge experienced (attraction experienced) by the valence (outer) electrons is greater in Be than in Li, so Be has a smaller atomic radius.

Chapter 03 Part 2: Intermolecular Forces

0	1	2 ✓

The student response earns two of the following points:

1 point is earned for indicating that Be has more protons than Li

1 point is earned for indicating that since the electrons are at about the same distance from the nucleus, there is more attraction in Be as a result of the larger number of protons

Both Li and Be have their outer electrons in the same shell (and/or they have the same number of inner core electrons shielding the valence electrons from the nucleus). However, Be has four protons and Li has only three protons. Therefore, the effective nuclear charge experienced (attraction experienced) by the valence (outer) electrons is greater in Be than in Li, so Be has a smaller atomic radius.

Part B

1 point is earned for saying that electrons are removed from an inner (third) level in potassium but one level higher, (fourth level) in calcium

1 point is earned for saying that the distance to the nucleus is less for the third level, so attraction is greater and more energy is needed to remove an electron

The second electron removed from a potassium atom comes from the third level (inner core). The second electron removed from a calcium atom comes from the fourth level (valence level). The electrons int he third level are closer to the nucleus so the attraction is much greater than for electrons in the fourth level.

0	1	2 ✓

The student response earns two of the following points:

1 point is earned for saying that electrons are removed from an inner (third) level in potassium but one level higher, (fourth level) in calcium

1 point is earned for saying that the distance to the nucleus is less for the third level, so attraction is greater and more energy is needed to remove an electron

The second electron removed from a potassium atom comes from the third level (inner core). The second electron removed from a calcium atom comes from the fourth level (valence level). The electrons int he third level are closer to the nucleus so the attraction is much greater than for electrons in the fourth level.

Part C

1 point is earned for indicating that C_2H_4 has a double bond and C_2H_6 has a single bond

Chapter 03 Part 2: Intermolecular Forces

1 point is earned for indicating that the carbon-carbon double bond in C_2H_4 required more energy to break (is stronger) than the carbon-carbon bond in C_2H_6

C_2H_4 has a double bond between the two carbon atoms, whereas C_2H_6 has a carbon-carbon single bond. More energy is required to break a double bond in C_2H_4 than to break a single bond in C_2H_6; therefore, the carbon-to-carbon bond energy in C_2H_4 is greater.

0	1	2

The student response earns two of the following points:

1 point is earned for indicating that C_2H_4 has a double bond and C_2H_6 has a single bond

1 point is earned for indicating that the carbon-carbon double bond in C_2H_4 required more energy to break (is stronger) than the carbon-carbon bond in C_2H_6

C_2H_4 has a double bond between the two carbon atoms, whereas C_2H_6 has a carbon-carbon single bond. More energy is required to break a double bond in C_2H_4 than to break a single bond in C_2H_6; therefore, the carbon-to-carbon bond energy in C_2H_4 is greater.

Part D

1 point is earned for indicating that Cl_2 and Br_2 are both nonpolar and/or have only London dispersion forces (or van der waals).

1 point for indicating that the more electrons, the more polarizable, the greater the dispersion forces, and the higher the boiling point.

C_2H_4 has a double bond between the two carbon atoms, whereas C_2H_6 has a carbon-carbon single bond. More energy is required to break a double bond in C_2H_4 than to break a single bond in C_2H_6; therefore, the carbon-to-carbon bond energy in C_2H_4 is greater.

0	1	2

The student response earns two of the following points:

1 point is earned for indicating that Cl_2 and Br_2 are both nonpolar and/or have only London dispersion forces (or van der waals).

1 point for indicating that the more electrons, the more polarizable, the greater the dispersion forces, and the higher the boiling point.

C_2H_4 has a double bond between the two carbon atoms, whereas C_2H_6 has a carbon-carbon single bond. More energy is required to break a double bond in C_2H_4 than to break a single bond in C_2H_6; therefore, the carbon-to-carbon bond energy in C_2H_4 is greater.

Chapter 03 Part 2: Intermolecular Forces

41. In which of the following liquids do the intermolecular forces include dipole-dipole forces?

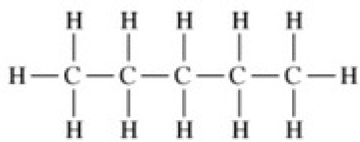
(A) $F_2(l)$

(B) $CH_4(l)$

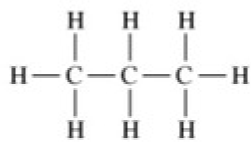
(C) $CF_4(l)$

(D) $CH_2F_2(l)$

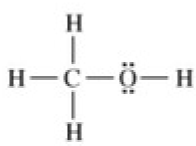
42. Answer the following questions in terms of principles of chemical bonding and intermolecular forces. In each explanation where a comparison is to be made, a complete answer must include a discussion of both substances. The following complete Lewis electron-dot diagrams may be useful in answering parts of this question.



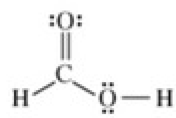
Pentane



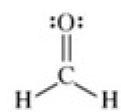
Propane



Methanol



Methanoic (formic) acid



Methanal (formaldehyde)

a. At 1 atm and 298 K, pentane is a liquid whereas propane is a gas. Explain.

b. At 1 atm and 298 K, methanol is a liquid whereas propane is a gas. Explain.

c. Indicate the hybridization of the carbon atom in each of the following:

   i. Methanol

   ii. Methanoic (formic) acid

d. Draw the complete Lewis electron-dot diagram for a molecule of propanoic acid, $HC_3H_5O_2$.

e. Explain the following observations about the two carbon-oxygen bonds in the methanoate (formate) anion, HCO_2^-. You may draw a Lewis electron-dot diagram (or diagrams) of the methanoate ion as part of your explanations.

   i. The two carbon-oxygen bonds in the methanoate (formate) anion, HCO_2^-, have the same length.

   ii. The length of the carbon-oxygen bonds in the methanoate (formate) anion, HCO_2^-, is intermediate between the length of the carbon-oxygen bond in methanol and the length of the carbon-oxygen bond in methanal.

Chapter 03 Part 2: Intermolecular Forces

Part A

1 point is earned for recognizing that pentane and propane *both* have only LDF's.

1 point is earned for recognizing that pentane has a larger electron cloud with greater IMF's and higher BP (reference to state of matter).

Both molecules are nonpolar, and the only intermolecular forces in each are London dispersion forces. However, pentane is larger than propane and has a more extensive electron cloud that can be involved in a greater number of London interactions, leading to stronger intermolecular attractions overall. Thus it takes a higher temperature for pentane molecules to have enough kinetic energy (on average) to overcome their stronger intermolecular attractions, thus pentane has the higher boiling point.

0	1	2 ✓

The student response earns two of the following points:

1 point is earned for recognizing that pentane and propane *both* have only LDF's.

1 point is earned for recognizing that pentane has a larger electron cloud with greater IMF's and higher BP (reference to state of matter).

Both molecules are nonpolar, and the only intermolecular forces in each are London dispersion forces. However, pentane is larger than propane and has a more extensive electron cloud that can be involved in a greater number of London interactions, leading to stronger intermolecular attractions overall. Thus it takes a higher temperature for pentane molecules to have enough kinetic energy (on average) to overcome their stronger intermolecular attractions, thus pentane has the higher boiling point.

Part B

1 point is earned for recognizing that methanol has hydrogen-bonding IMF's and propane has LDF's.

1 point is earned for recognizing that hydrogen-bonding results in a greater IMF causing methanol to be a liquid.

Propane molecules are nonpolar and only interact with one another via London dispersion forces. Methanol molecules are polar and hydrogen bonds (as well as London forces) can form among them. Because hydrogen bonds are stronger than London forces, methanol has greater intermolecular attractions. Thus it takes a higher temperature for methanol molecules to have enough kinetic energy (on average) to overcome their stronger intermolecular attractions, thus methanol has the higher boiling point.

0	1	2 ✓

The student response earns two of the following points:

1 point is earned for recognizing that methanol has hydrogen-bonding IMF's and propane has LDF's.

Chapter 03 Part 2: Intermolecular Forces

1 point is earned for recognizing that hydrogen-bonding results in a greater IMF causing methanol to be a liquid.

Propane molecules are nonpolar and only interact with one another via London dispersion forces. Methanol molecules are polar and hydrogen bonds (as well as London forces) can form among them. Because hydrogen bonds are stronger than London forces, methanol has greater intermolecular attractions. Thus it takes a higher temperature for methanol molecules to have enough kinetic energy (on average) to overcome their stronger intermolecular attractions, thus methanol has the higher boiling point.

Part C

- 1 point is earned for the correct hybridization.

  *sp*3

- 1 point is earned for the correct hybridization.

  *sp*2

0	1	2

The student response earns two of the following points:

- 1 point is earned for the correct hybridization.

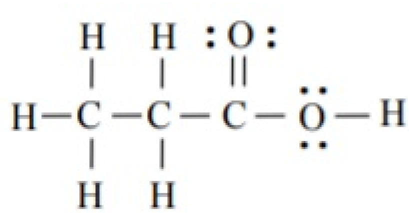
  *sp*3

- 1 point is earned for the correct hybridization.

  *sp*2

Part D

1 point is earned for the correct diagram (all electron pairs must be included).

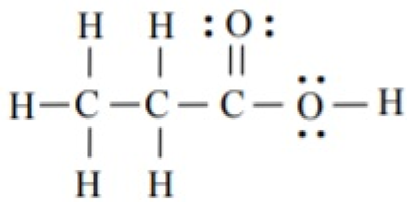


0	1

The student response earns one of the following points:

Chapter 03 Part 2: Intermolecular Forces

1 point is earned for the correct diagram (all electron pairs must be included).



Part E

- 1 point is earned for the correct explanation.

  Resonance structures can be drawn for the methanoate anion, with one carbonoxygen bond as a single bond and the other as a double bond (and the opposite in the other structure). The electrons are shared equally between the two carbon-oxygen bonds in the methanoate anion, resulting in two bonds with the same length.

- 1 point is earned for the correct explanation comparing the single, double and resonating bonds of the respective molecules.

  The two identical carbon-oxygen bonds in the methanoate anion each have a bond order of 1.5. The carbon-oxygen bond in methanol is a single bond, and the carbon-oxygen bond in methanal is a double bond. Single bonds between atoms of the same two elements are longer than double bonds, and a bond with a bond order of 1.5 would have a bond length between that of a single bond and that of a double bond.

0	1	2

The student response earns two of the following points:

- 1 point is earned for the correct explanation.

  Resonance structures can be drawn for the methanoate anion, with one carbonoxygen bond as a single bond and the other as a double bond (and the opposite in the other structure). The electrons are shared equally between the two carbon-oxygen bonds in the methanoate anion, resulting in two bonds with the same length.

- 1 point is earned for the correct explanation comparing the single, double and resonating bonds of the respective molecules.

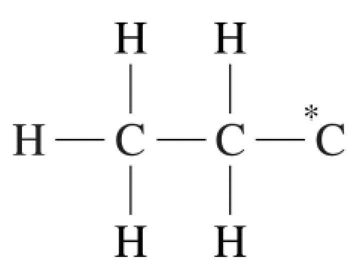
  The two identical carbon-oxygen bonds in the methanoate anion each have a bond order of 1.5. The carbon-oxygen bond in methanol is a single bond, and the carbon-oxygen bond in methanal is a double bond. Single bonds between atoms of the same two elements are longer than double bonds, and a bond with a bond order of 1.5 would have a bond length between that of a single bond and that of a double bond.

Chapter 03 Part 2: Intermolecular Forces

43. **Directions:** Examples and equations may be included in your responses where appropriate. For calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Pay attention to significant figures

Propanoic acid, C2H5COOH, is an organic acid that is a liquid at room temperature.

(a) An incomplete Lewis diagram for the propanoic acid molecule is provided in the box below. Complete the diagram, showing how the remaining atoms in the molecule are arranged around the carbon atom marked with an asterisk (*). Your structure should minimize formal charge and include any lone pairs of electrons.



(b) Identify the hybridization of the carbon atom marked with the asterisk

(c) Propanoic acid has a lower boiling point than butanoic acid, C_3H_7COOH.

i. Identify all the types of intermolecular forces present among the molecules in propanoic acid.

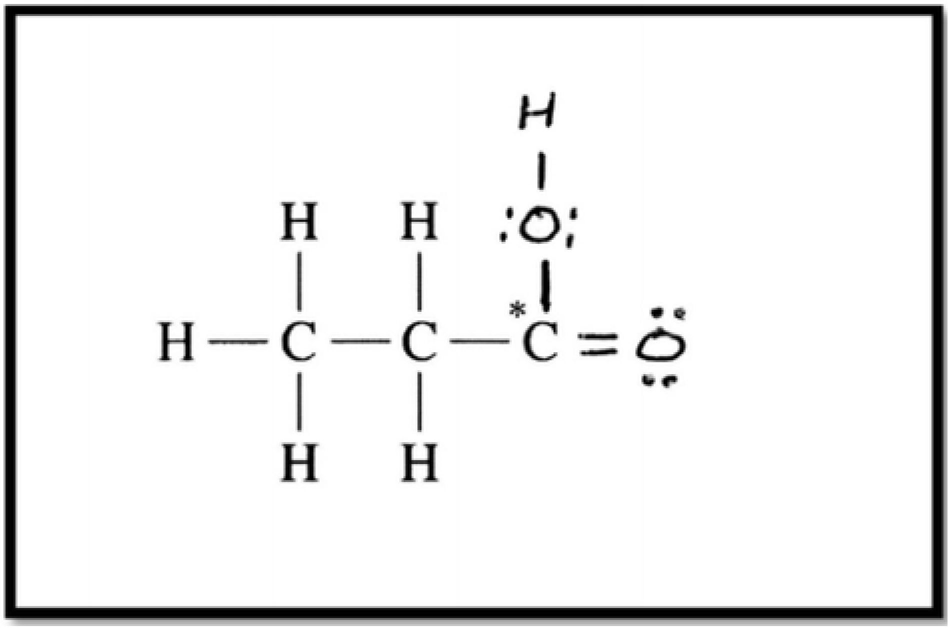
ii. Which of the types of intermolecular forces that you identified in part (c)(i) is most responsible for the difference in boiling points of the two acids?

Part A

1 point(s) maximum

1 point is earned for a correct structure.

Chapter 03 Part 2: Intermolecular Forces



There should be two O atoms attached to the C atom, one with a double bond and one with a single bond. The O atom attached with a single bond should have an H atom attached to it. Each O atom has two lone pairs of electrons. (Figure is required.)

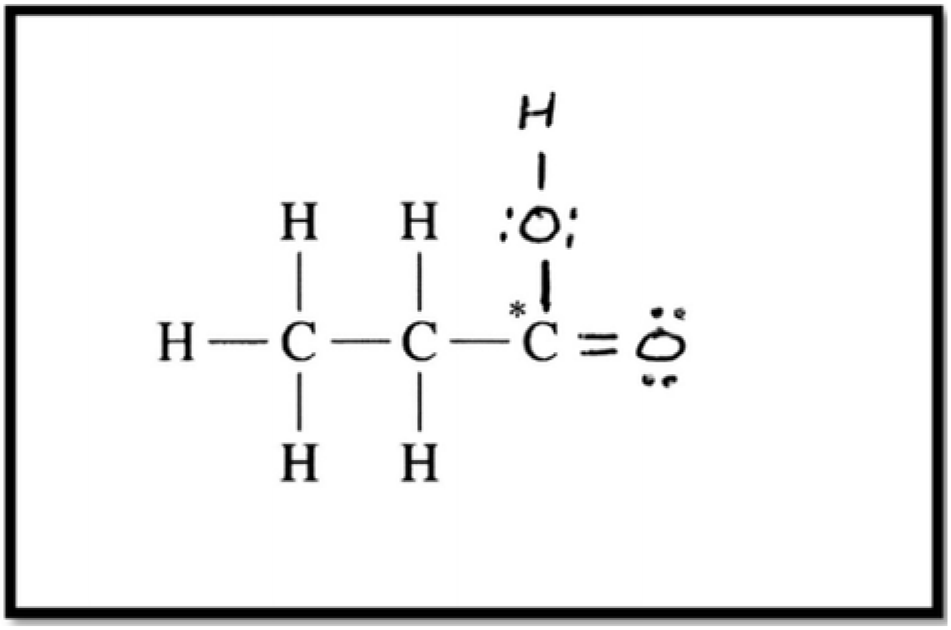
0	1

Student response earns 1 of the following 1 point(s)

1 point(s) maximum

1 point is earned for a correct structure.

Chapter 03 Part 2: Intermolecular Forces



There should be two O atoms attached to the C atom, one with a double bond and one with a single bond. The O atom attached with a single bond should have an H atom attached to it. Each O atom has two lone pairs of electrons. (Figure is required.)

Part B

1 point(s) maximum

1 point is earned for the correct hybridization.

sp2

0	1

Student response earns 1 of the following 1 point(s)

1 point(s) maximum

1 point is earned for the correct hybridization.

sp2

Part C

2 point(s) maximum

Part (i)

1 point is earned for identifying both London dispersion forces and hydrogen bonding.

Chapter 03 Part 2: Intermolecular Forces

London dispersion forces, dipole-dipole forces, and hydrogen bonding. (Identifying dipole-dipole forces is not required to earn the point.)

Part (ii)

1 point is earned for the correct answer.

London dispersion forces.

0	1	2

Student response earns 2 of the following 2 point(s)

2 point(s) maximum

Part (i)

1 point is earned for identifying both London dispersion forces and hydrogen bonding.

London dispersion forces, dipole-dipole forces, and hydrogen bonding. (Identifying dipole-dipole forces is not required to earn the point.)

Part (ii)

1 point is earned for the correct answer.

London dispersion forces.

Chapter 03 Part 2: Intermolecular Forces

44. Answer the following questions related to sulfur and one of its compounds.

a. Consider the two chemical species S and S^{2-} .

i. Write the electron configuration (e.g., $1s^2\ 2s^2$ . . .) of each species.

ii. Explain why the radius of the S^{2-} ion is larger than the radius of the S atom.

iii. Which of the two species would be attracted into a magnetic field? Explain.

b. The S^{2-} ion is isoelectronic with the Ar atom. From which species, S^{2-} or Ar, is it easier to remove an electron? Explain.

c. In the H_2S molecule, the H–S–H bond angle is close to 90°. On the basis of this information, which atomic orbitals of the S atom are involved in bonding with the H atoms?

d. Two types of intermolecular forces present in liquid H_2S are London (dispersion) forces and dipole-dipole forces.

i. Compare the strength of the London (dispersion) forces in liquid H_2S to the strength of the London (dispersion) forces in liquid H_2O. Explain.

ii. Compare the strength of the dipole-dipole forces in liquid H_2S to the strength of the dipole-dipole forces in liquid H_2O. Explain.

Part A

- 1 point is earned for the correct configuration for S.

S: 1*s*2 2*s*2 2*p*6 3*s*2 3*p*4

1 point is earned for the correct configuration for S2–.

S2−: 1*s*2 2*s*2 2*p*6 3*s*2 3*p*6

Note: Replacement of 1*s*2 2*s*2 2*p*6 by [Ne] is acceptable.

- 1 point is earned for a correct explanation.

The nuclear charge is the same for both species, but the eight valence electrons in the sulfide ion experience a greater amount of electron-electron repulsion than do the six valence electrons in the neutral sulfur atom. This extra repulsion in the sulfide ion increases the average distance between the valence electrons, so the electron cloud around the sulfide ion has the greater radius.

- 1 point is earned for the correct answer with a correct explanation.

The sulfur atom would be attracted into a magnetic field. Sulfur has two unpaired *p* electrons, which results in a net magnetic moment for the atom. This net magnetic moment would interact with an external magnetic field, causing a net attraction into the field. The sulfide ion would not be attracted into a magnetic field because all the electrons in the species are paired, meaning that their individual magnetic moments would cancel each other.

Chapter 03 Part 2: Intermediate Forces

0	1	2	3	4

The student response earns four of the following points:

- 1 point is earned for the correct configuration for S.

  S: 1*s*2 2*s*2 2*p*6 3*s*2 3*p*4

  1 point is earned for the correct configuration for S2–.

  S2−: 1*s*2 2*s*2 2*p*6 3*s*2 3*p*6

  Note: Replacement of 1*s*2 2*s*2 2*p*6 by [Ne] is acceptable.

- 1 point is earned for a correct explanation.

  The nuclear charge is the same for both species, but the eight valence electrons in the sulfide ion experience a greater amount of electron-electron repulsion than do the six valence electrons in the neutral sulfur atom. This extra repulsion in the sulfide ion increases the average distance between the valence electrons, so the electron cloud around the sulfide ion has the greater radius.

- 1 point is earned for the correct answer with a correct explanation.

  The sulfur atom would be attracted into a magnetic field. Sulfur has two unpaired *p* electrons, which results in a net magnetic moment for the atom. This net magnetic moment would interact with an external magnetic field, causing a net attraction into the field. The sulfide ion would not be attracted into a magnetic field because all the electrons in the species are paired, meaning that their individual magnetic moments would cancel each other.

Part B

1 point is earned for the correct answer with a correct explanation.

It requires less energy to remove an electron from a sulfide ion than from an argon atom. A valence electron in the sulfide ion is less attracted to the nucleus (charge +16) than is a valence electron in the argon atom (charge +18).

0	1

The student response earns one of the following points:

1 point is earned for the correct answer with a correct explanation.

It requires less energy to remove an electron from a sulfide ion than from an argon atom. A valence electron in the sulfide ion is less attracted to the nucleus (charge +16) than is a valence electron in the argon atom (charge +18).

Chapter 03 Part 2: Intermolecular Forces

Part C

1 point is earned for the correct answer.

The atomic orbitals involved in bonding with the H atoms in $H2S$ are *p* (specifically, 3*p*) orbitals. The three *p* orbitals are mutually perpendicular (i.e., at 90°) to one another.

0	1

The student response earns one of the following points:

1 point is earned for the correct answer.

The atomic orbitals involved in bonding with the H atoms in $H2S$ are *p* (specifically, 3*p*) orbitals. The three *p* orbitals are mutually perpendicular (i.e., at 90°) to one another.

Part D

- 1 point is earned for the correct answer with a correct explanation.

  The strength of the London forces in liquid $H2S$ is greater than that of the London forces in liquid $H2O$. The electron cloud of $H2S$ has more electrons and is thus more polarizable than the electron cloud of the $H2O$ molecule.

- 1 point is earned for the correct answer with a correct explanation.

  The strength of the dipole-dipole forces in liquid $H2S$ is weaker than that of the dipole-dipole forces in liquid $H2O$. The net dipole moment of the $H2S$ molecule is less than that of the $H2O$ molecule. This results from the lesser polarity of the H–S bond compared with that of the H–O bond (S is less electronegative than O).

0	1	2

The student response earns two of the following points:

- 1 point is earned for the correct answer with a correct explanation.

  The strength of the London forces in liquid $H2S$ is greater than that of the London forces in liquid $H2O$. The electron cloud of $H2S$ has more electrons and is thus more polarizable than the electron cloud of the $H2O$ molecule.

- 1 point is earned for the correct answer with a correct explanation.

  The strength of the dipole-dipole forces in liquid $H2S$ is weaker than that of the dipole-dipole forces in liquid $H2O$. The net dipole moment of the $H2S$ molecule is less than that of the $H2O$ molecule. This results from the lesser polarity of the H–S bond compared with that of the H–O bond (S is less electronegative than O).